

Final Project

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2024-02-27

Task 0

- Files that do not follow Task 0 will get points deducted from the top of the assignment.
- Complete the following tasks and ensure your completed code to each question is copied into this .Rmd file into the appropriate R chunk, and that your final file is knitted into a .pdf document as already set up in this document. Leave the labels and instructions for each task. Include your name and date under the author and data portion above in quotations.
- Important: To avoid point reductions:
 - Do not include code that you don't need outside of the assignment. This includes not printing out any complete data frames or data sets and variable values that are not explicitly asked for.
 - Ensure your code requested to generate an answer is executable and not in comments.
 - Proof read your submission to make sure you have included all files.
 - Ensure sure your assessment has been checked for readability.
 - Knit this assessment from a .Rmd file to .pdf file.
 - Include required libraries and links to datasets (read.csv) where necessary.
 - Always refer to file locations, paths, or directories relatively by putting a datasets in a data folder in your working directory and using "data/...csv" inside any read data lines of code. This means there should be no setwd() commands inside your .Rmd file.
 - Do not alter any data sets at all (e.g. remove extraneous rows or columns, change certain values manually, or rename files) prior to uploading in R.
 - Refrain from including commands that generate output that spans multiple pages and leads to substantial scrolling of the pdf.

Resources Required

- Spend a few minutes looking through the documents on Canvas and <https://www.realtor.com/research/data/> before you begin. For this Final Project, you may use historical data at the state level, metro level, county level or zip level under the inventory - monthly category. Data and a data dictionary are provided to you on Canvas.
- Choose one of the following:
 - RDC_Inventory_Core_Metrics_County_History.csv,
 - RDC_Inventory_Core_Metrics_Metro_History.csv,
 - RDC_Inventory_Core_Metrics_State_History.csv,
 - RDC_Inventory_Core_Metrics_Zip_History.csv

Overall Goal

- Your goal is to examine the relationships with Price in the housing market. In preparing the data, you can create subsets of the data set to make it easier to work with based on what you need. Read all the instructions below before making decisions on how to slice down the data set. Take note that some of the variables are measuring a similar thing, so be sure to choose unique variables to have the most accurate results.

Task 1 (25 points)

- Take steps to read in and prepare the data, ensuring that there are no extra lines that should not be in the dataset and that the variables you choose are in the correct data types.
- Save a new smaller data object that only includes the variables you choose to work with throughout the project.
- If you want to eliminate rows by subsetting a smaller dataset after eliminating columns in the step above, do so here and provide the rationale based on what you want to examine.
- Code any missing values appropriately and rename variables where necessary with explanation in comments of what the new named variable represents.
- Comment on how you cleaned the data and your rationale for doing so.

```
# Load libraries
library(tidyverse)
library(semTools)
library("car")

# Load dataset
county.data <- read.csv(file = "data/RDC_Inventory_Core_Metrics_County_History.csv")

# Clean data
county.cleaned <- county.data %>%
  ## Select variables
  select(month_date_YYYYMM, county_name, median_listing_price, average_listing_price,
    active_listing_count, median_days_on_market, quality_flag) %>%
  ## Drop NAs for average_listing_price
  drop_na(average_listing_price) %>%
  ## Remove '1' values in quality_flag
  mutate(quality_flag = coalesce(quality_flag, 0)) %>%
  mutate(quality_flag = na_if(x = quality_flag, y = 1)) %>%
  drop_na(quality_flag) %>%
  ## Separate county_name into 'county' and 'state'
  separate_wider_delim(county_name, delim = ",", names = c("county", "state")) %>%
  ## Split month_date_YYYYMM column into 'year' and 'month'
  separate_wider_position(month_date_YYYYMM, c(year = 4, month = 2), cols_remove = FALSE) %>%
  mutate(year = as.factor(year)) %>%
  mutate(month = as.factor(month)) %>%
  mutate(month = recode_factor(.x = month, `01` = "Jan", `02` = "Feb",
    `03` = "Mar", `04` = "Apr", `05` = "May", `06` = "Jun", `07` = "Jul",
    `08` = "Aug", `09` = "Sep", `10` = "Oct", `11` = "Nov", `12` = "Dec"))

# Filter data to only include Virginia cities of interest. Recode
# variables.
va.cleaned <- county.cleaned %>%
```

```
filter(state == "va") %>%
filter(county %in% c("virginia beach city", "norfolk city", "chesapeake city",
  "portsmouth city", "suffolk city")) %>%
mutate(county = as.factor(county)) %>%
mutate(state = as.factor(state)) %>%
mutate(month_date_yyyymm = as.numeric(month_date_yyyymm))
```

Task 2 (30 points)

- Examine at least 6 variables from the data set including all measures of central tendency and spread that we covered in the course for continuous variables, and frequency and relative frequency for categorical variables. Make your output clear by numbering your variables 1 - 6 and outputting all summary measures at ones.
- Depending on how you sliced the data, you may only have one categorical variable out of the 6.
- Provide rationale and insights on what you took away from this step in comments.

```
# 1. Year - Frequency and Relative Frequency. Examining # of
# observations per year. Fewer observations in 2016 and 2022 when
# compared to the other years.
(year.freq <- table(va.cleaned$year))
```

```
##
## 2016 2017 2018 2019 2020 2021 2022
##    30   60   60   60   59   59   40
```

```
prop.table(year.freq)
```

```
##
##      2016      2017      2018      2019      2020      2021      2022
## 0.08152174 0.16304348 0.16304348 0.16304348 0.16032609 0.16032609 0.10869565
```

```
# 2. County - Frequency and Relative Frequency. Examining # of
# observations per county. Observations for each counties equals 74,
# except Virginia Beach (72).
(county.freq <- table(va.cleaned$county))
```

```
##
##      chesapeake city      norfolk city      portsmouth city      suffolk city
##              74              74              74              74
## virginia beach city
##              72
```

```
prop.table(county.freq)
```

```
##
##      chesapeake city      norfolk city      portsmouth city      suffolk city
##      0.2010870      0.2010870      0.2010870      0.2010870
## virginia beach city
##      0.1956522
```

```
# 3. Median Listing Price - Large sample size: -7 to 7. Distribution
# is left skewed and platykurtic, but not problematic, [-7 < z < 7].
```

```
va.cleaned %>%
  summarize(mean.mlp = mean(median_listing_price), med.mlp = median(median_listing_price),
            sd.mlp = sd(median_listing_price), var.mlp = var(median_listing_price),
            IQR.mlp = IQR(median_listing_price), mode.mlp = names(x = sort(x = table(median_listing_price),
            decreasing = TRUE))[1])
```

```
## # A tibble: 1 x 6
##   mean.mlp med.mlp sd.mlp      var.mlp IQR.mlp mode.mlp
##   <dbl>    <dbl> <dbl>      <dbl>  <dbl> <chr>
## 1  282882.  299900 69874. 4882390018. 115516 299900
```

```
range(va.cleaned$median_listing_price)
```

```
## [1] 144200 415680
```

```
skew(va.cleaned$median_listing_price)
```

```
## skew (g1)          se          z          p
## -0.3850500  0.1276885 -3.0155425  0.0025652
```

```
kurtosis(va.cleaned$median_listing_price)
```

```
## Excess Kur (g2)          se          z          p
## -0.8853355229  0.2553769592  -3.4667791704  0.0005267344
```

```
# 4. Average Listing Price - Large sample size: -7 to 7. Distribution
# is right skewed and platykurtic, but not problematic, [-7 < z < 7].
```

```
va.cleaned %>%
  summarize(mean.alp = mean(average_listing_price), med.alp = median(average_listing_price),
            sd.alp = sd(average_listing_price), var.alp = var(average_listing_price),
            IQR.alp = IQR(average_listing_price), mode.alp = names(x = sort(x = table(average_listing_price),
            decreasing = TRUE))[1])
```

```
## # A tibble: 1 x 6
##   mean.alp med.alp sd.alp      var.alp IQR.alp mode.alp
##   <dbl>    <dbl> <dbl>      <dbl>  <dbl> <chr>
## 1  334422.  326882. 100530. 10106277204. 133704 159916
```

```
range(va.cleaned$average_listing_price)
```

```
## [1] 159916 590870
```

```
skew(va.cleaned$average_listing_price)
```

```
## skew (g1)          se          z          p
## 0.25145056 0.12768848 1.96925020 0.04892437
```

```
kurtosis(va.cleaned$average_listing_price)
```

```
## Excess Kur (g2)          se          z          p
##      -0.3250993      0.2553770     -1.2730173     0.2030119
```

```
# 5. Median Days on Market - Large sample size: -7 to 7. Distribution
# is right skewed and platykurtic, but not problematic, [-7 < z < 7].
```

```
va.cleaned %>%
```

```
  summarize(mean.dom = mean(median_days_on_market), med.dom = median(median_days_on_market),
            sd.dom = sd(median_days_on_market), var.dom = var(median_days_on_market),
            IQR.dom = IQR(median_days_on_market), mode.dom = names(x = sort(x = table(median_days_on_market,
            decreasing = TRUE)))[1])
```

```
## # A tibble: 1 x 6
```

```
##   mean.dom med.dom sd.dom var.dom IQR.dom mode.dom
##   <dbl>   <dbl> <dbl>   <dbl>   <dbl>   <chr>
## 1    46.8     47  17.8   316.    28.2  25
```

```
range(va.cleaned$median_days_on_market)
```

```
## [1] 13 89
```

```
skew(va.cleaned$median_days_on_market)
```

```
## skew (g1)          se          z          p
## 0.1102565 0.1276885 0.8634800 0.3878736
```

```
kurtosis(va.cleaned$median_days_on_market)
```

```
## Excess Kur (g2)          se          z          p
##      -0.818521574      0.255376959     -3.205150445     0.001349919
```

```
# 6. Active Listing Count - Large sample size: -7 to 7. Distribution
# is right skewed to a problematic degree, [z > 7]. Distribution is
# leptokurtic, but not problematic, [-7 < z < 7].
```

```
va.cleaned %>%
```

```
  summarize(mean.alc = mean(active_listing_count), med.alc = median(active_listing_count),
            sd.alc = sd(active_listing_count), var.alc = var(active_listing_count),
            IQR.alc = IQR(active_listing_count), mode.alc = names(x = sort(x = table(active_listing_count,
            decreasing = TRUE)))[1])
```

```
## # A tibble: 1 x 6
```

```
##   mean.alc med.alc sd.alc var.alc IQR.alc mode.alc
##   <dbl>   <dbl> <dbl>   <dbl>   <dbl>   <chr>
## 1    887.    684  592. 350361.    758 302
```

```
range(va.cleaned$active_listing_count)
```

```
## [1] 135 2582
```

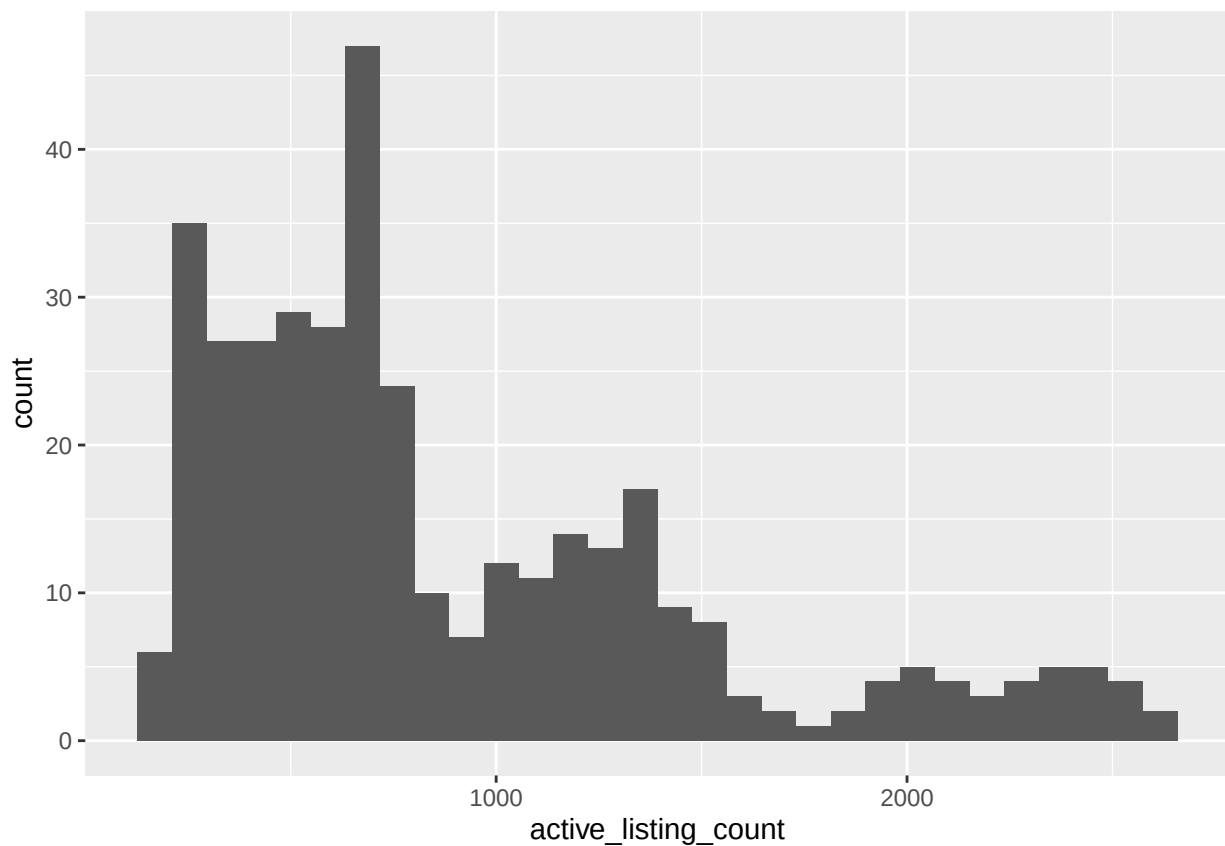
```
skew(va.cleaned$active_listing_count)
```

```
## skew (g1)      se      z      p
## 1.1867363 0.1276885 9.2939965 0.0000000
```

```
kurtosis(va.cleaned$active_listing_count)
```

```
## Excess Kur (g2)      se      z      p
## 0.720023887 0.255376959 2.819455166 0.004810525
```

```
va.cleaned %>%  
  ggplot(aes(active_listing_count)) + geom_histogram()
```



Task 3 (30 points)

- Use at least 3 grouping functions to group the data in order to see data at different levels. You may group categorical variables further to see different patterns based on location or even groups of dates. Comment on the insights you took away from this step.

```
# Group by Year/Average Listing Price. Average Listing Price has  
# consistently increased each year.
```

```

year.alp <- va.cleaned %>%
  group_by(year) %>%
  summarise(average_listing_price = mean(average_listing_price))

# Group by Year/Active Listing Count. Active Listing Count has
# consistently decreased each year.
year.alc <- va.cleaned %>%
  group_by(year) %>%
  summarise(active_listing_count = mean(active_listing_count))

# Group by Year/County/Average Listing Price. Virginia Beach has
# consistently higher average listing prices than the other counties.
# Portsmouth has consistently lower average listing prices than the
# other counties.
year.county.alp <- va.cleaned %>%
  group_by(year, county) %>%
  summarise(average_listing_price = mean(average_listing_price))

# Group by Year/County/Active Listing Count. Virginia Beach has
# consistently higher active listing counts than the other counties.
# This might suggest that location has a significant impact on
# average listing price, regardless of active listing count.
year.county.alc <- va.cleaned %>%
  group_by(year, county) %>%
  summarise(active_listing_count = mean(active_listing_count))

```

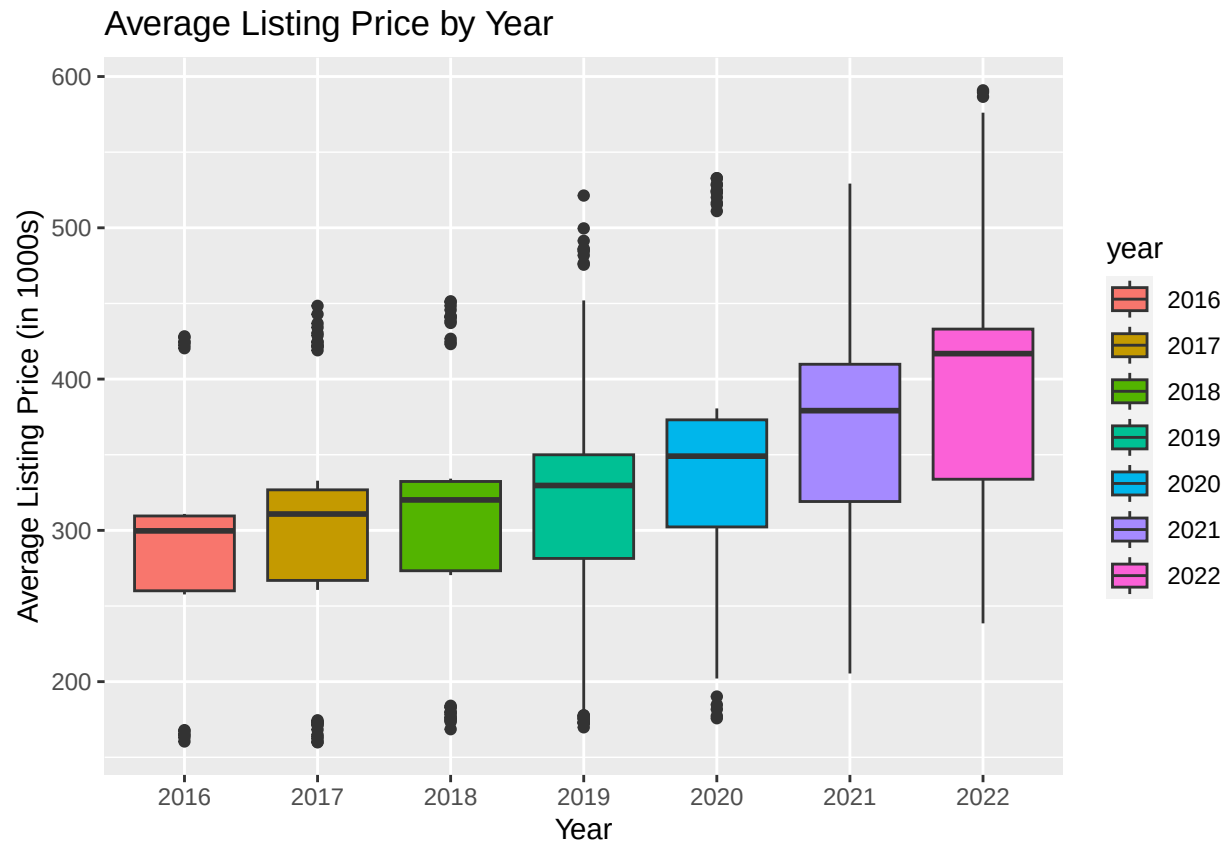
Task 4 (40 points)

- Create at least 4 well-formatted, appropriate graphs depicting relationships between 2 variables (used or created in the step above) of your choosing. Use good code-formatting practices and ensure your graphs have good use of parameters including color, labels, legends, or titles - where appropriate. Include a sentence or two after each graph in comments that explains what the graph shows and what you learned from the graph.

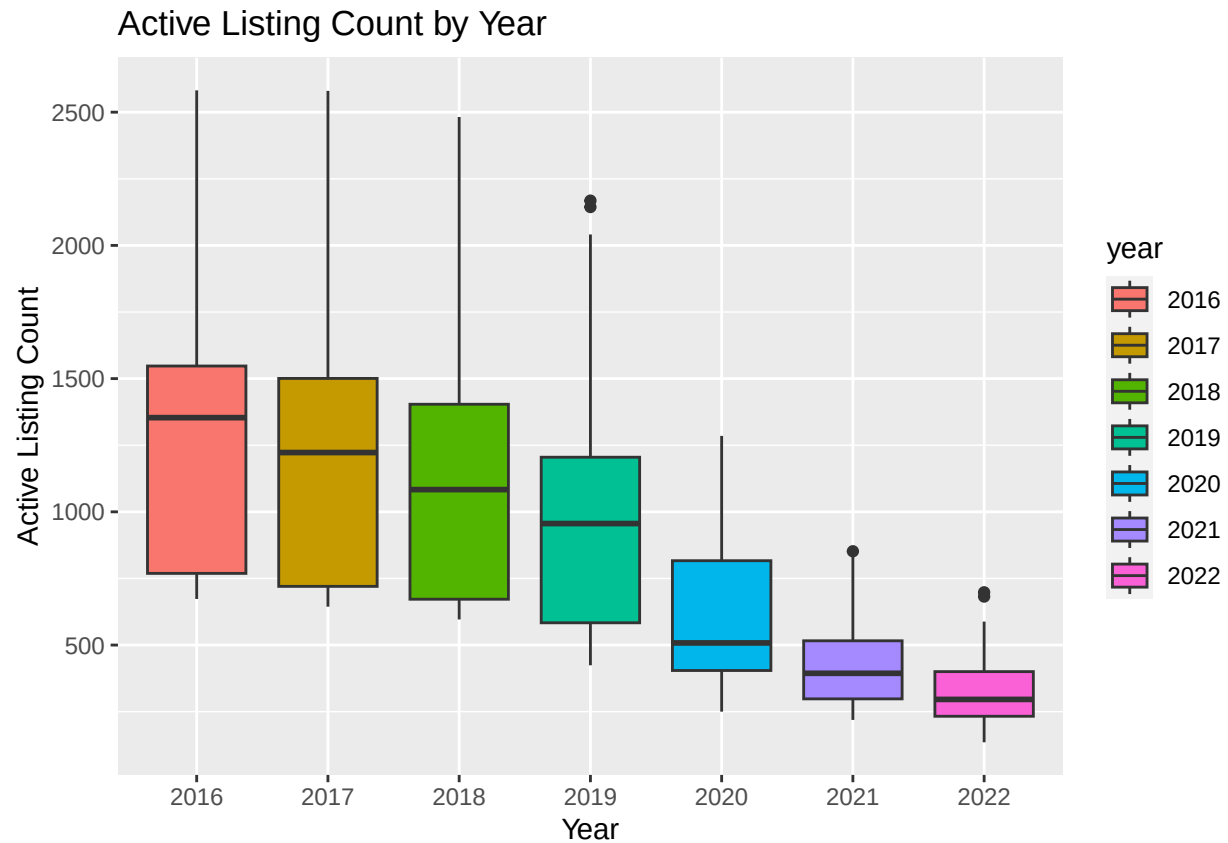
```

# Boxplot of Average Listing Price by Year. Average Listing Price has
# increased every year from 2016 to 2022. Several outliers present.
(plot.year.alp <- va.cleaned %>%
  ggplot(aes(x = year, y = average_listing_price/1000)) + geom_boxplot(aes(fill = year)) +
  labs(x = "Year", y = "Average Listing Price (in 1000s)") + ggtitle("Average Listing Price by Year"))

```



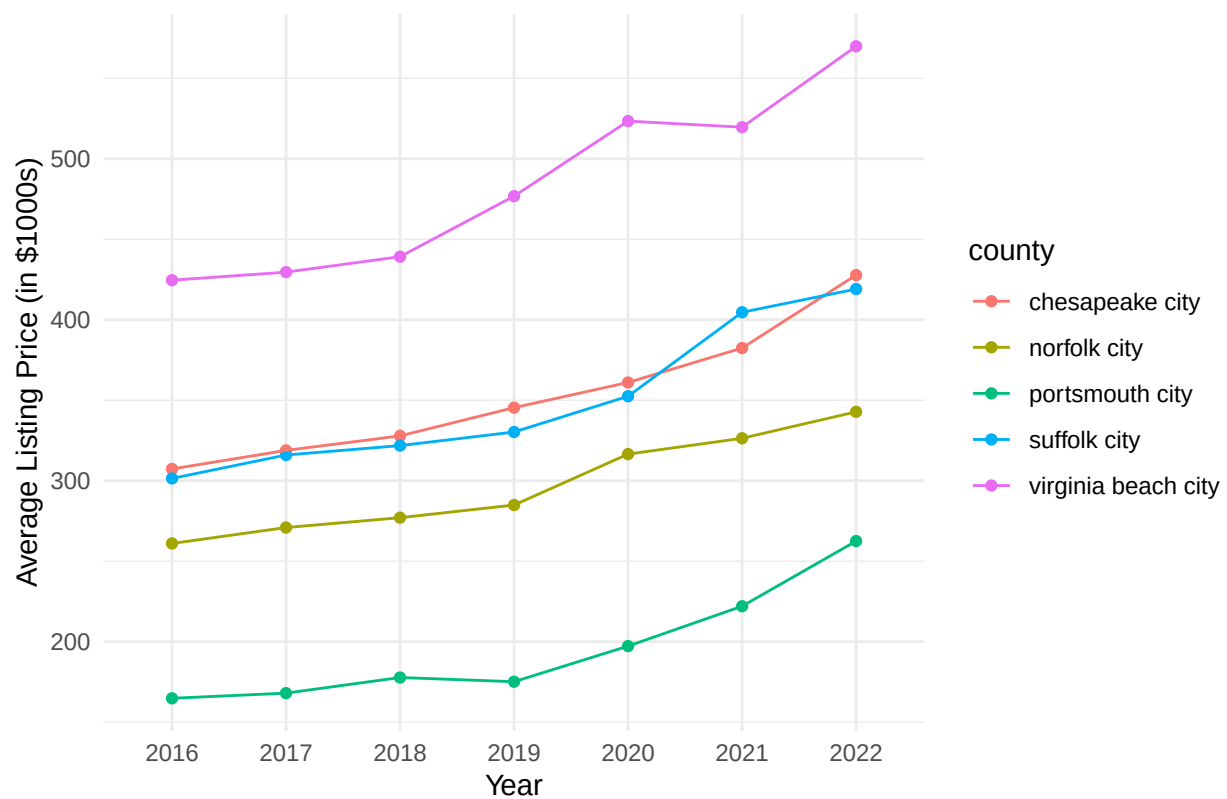
```
# Boxplot of Active Listing Count by Year. Active Listing Count has
# decreased every year from 2016 to 2022. Outliers present in 2019,
# 2021, and 2022.
(plot.year.alc <- va.cleaned %>%
  ggplot(aes(x = year, y = active_listing_count)) + geom_boxplot(aes(fill = year)) +
  labs(x = "Year", y = "Active Listing Count") + ggtitle("Active Listing Count by Year"))
```

```
# Plot of Average Listing Price (in $1000s) by County between 2016 -
# 2022. Average Listing Price has consistently increased each year
# for all selected counties. However, Average Listing Price does
# differ depending on the county. For example, Virginia Beach is
# consistently more expensive than the other cities, regardless of
# year. Portsmouth is consistently less expensive than the other
# counties, regardless of year.
```

```
(point.alp.year <- year.county.alp %>%
  ggplot(aes(x = year, y = average_listing_price/1000)) + geom_point(aes(color = county)) +
  geom_line(aes(color = county, group = county)) + theme_minimal() +
  labs(x = "Year", y = "Average Listing Price (in $1000s)") + ggtitle("Average Listing Price by County"))
```

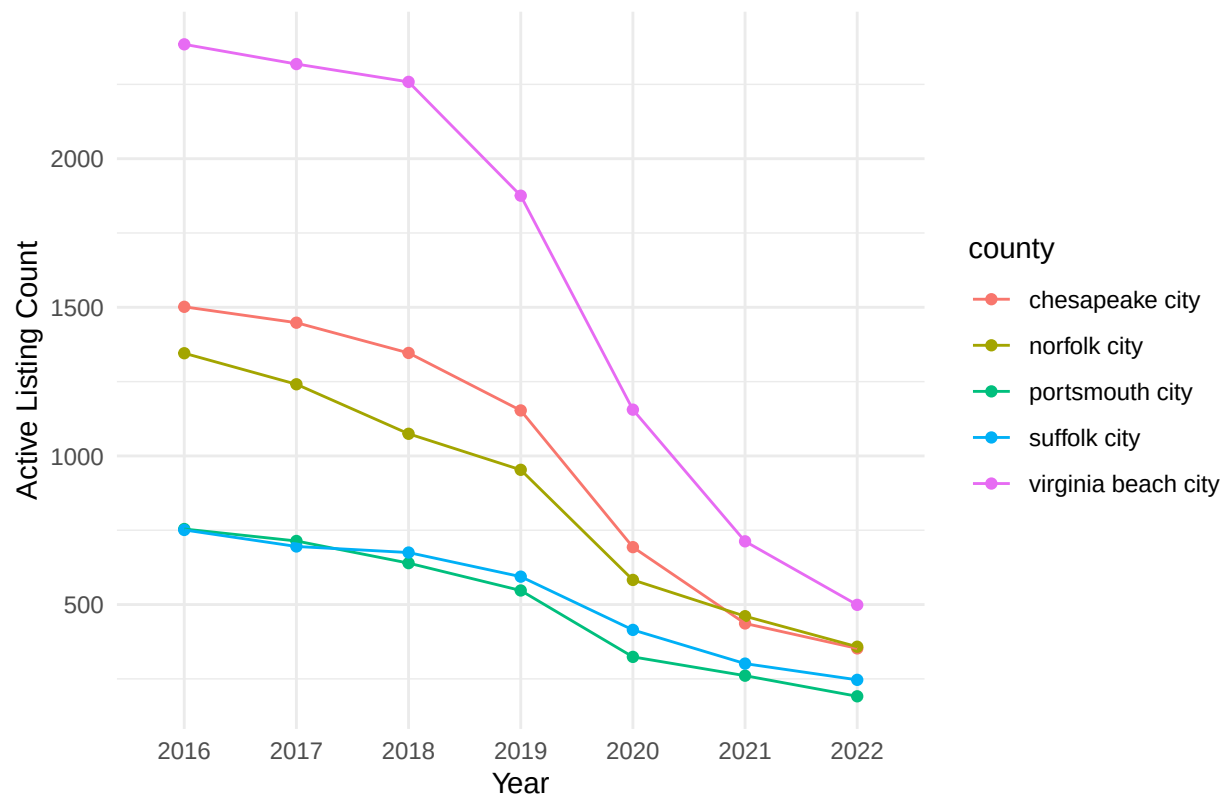
Average Listing Price by County between 2016–2022



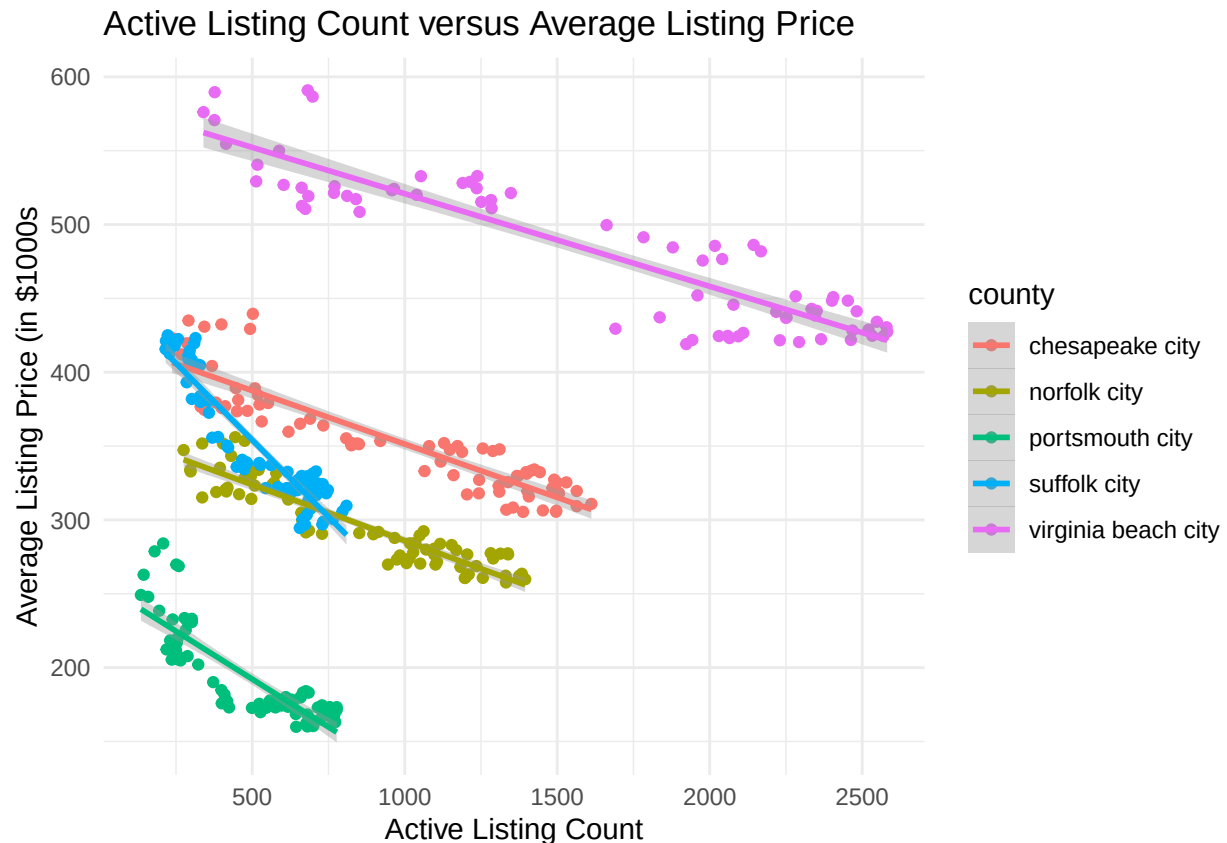
Plot of Active Listing Count by County between 2016 - 2022. Active Listing Count has consistently decreased each year for all selected counties. However, the slope of decrease seems to vary by county and year.

```
(point.alc.year <- year.county.alc %>%
  ggplot(aes(x = year, y = active_listing_count)) + geom_point(aes(color = county)) +
  geom_line(aes(color = county, group = county)) + geom_smooth() + theme_minimal() +
  labs(x = "Year", y = "Active Listing Count") + ggtitle("Active Listing Count by County between 2016 - 2022"))
```

Active Listing Count by County between 2016–2022



```
# Plot of Average Listing Price (in $1000s) versus Active Listing
# Count, by County. For each county, Active Listing Price decreases
# as Active Listing Count increases.
(plot.alc.alp.county <- va.cleaned %>%
  ggplot(aes(x = active_listing_count, y = average_listing_price/1000,
    color = county)) + geom_point() + geom_smooth(method = "lm") +
  theme_minimal() + labs(x = "Active Listing Count", y = "Average Listing Price (in $1000s)" +
    ggtitle("Active Listing Count versus Average Listing Price"))
```



Task 5 (20 points)

- Make some general observations on what you would find when you compare variables based on what you found above in the data.

Average Listing Price has consistently increased each year, and
 # across all counties of interest. Outliers present in all years,
 # except 2021. I assume outliers are due to large disparities based
 # on location/county, particularly between Virginia Beach and
 # Portsmouth.

Active Listing Count has consistently decreased each year, and
 # across all counties of interest. Outliers present in 2019, 2021,
 # and 2022. Additionally, there seemed to be a more dramatic decrease
 # in average Active Listing Counts between 2019-2022, particularly
 # for Virginia Beach, than prior years.

Average Listing Price differs across counties of interest. Virginia
 # Beach is consistently the most expensive, with the highest Average
 # Listing Prices between 2016-2022. Chesapeake and Suffolk are
 # similarly priced between 2016-2022. Norfolk has the 2nd lowest
 # Average Listing Prices between 2016-2022. Lastly, Portsmouth is
 # consistently the least expensive, with the lowest Average Listing
 # Prices across 2016-2022.

```
# Active Listing Count differs across counties. Virginia Beach
# consistently has the highest average Active Listing Count each
# year. Followed by Norfolk and Chesapeake, which have similar
# average Active Listing Counts across 2016-2022. Lastly, Portsmouth
# and Suffolk have the lowest Active Listing Counts across 2016-2022.
```

Task 6 (60 points)

- Select the appropriate test to test at least 4 relationships between a Price variable as a Y variable (median or average, percentage or otherwise) and at least one other variable. Do not to choose another price variable in making predictions of price. Go through the correct steps for hypothesis testing based on the test you selected including listing your conclusions you made and whether they differ from your predictions above in Step 5 after examining relationships visually.
- You may choose to use ANOVA, simple, or multiple regression to evaluate 4 relationships with regards to Y to satisfy the requirements. If you choose an ANOVA, be sure to conduct appropriate Post Hoc Tests.

```
# Test 1 & 2: Differences in Average Price Listing across Years and
# Counties. Step 1:

# Hypothesis 1:

# H0 - The Average Listing Price does not differ across years
# (2016-2022).

# HA - The Average Listing Price does differ across years
# (2016-2022).

# Hypothesis 2:

# H0 - The Average Listing Price does not differ across counties.

# HA - The Average Listing Price does differ across counties.

# Step 2 & 3: Compute test-statistic.

aov.alp.year.county <- aov(average_listing_price ~ year + county, data = va.cleaned)
summary(aov.alp.year.county)
```

```
##           Df    Sum Sq  Mean Sq F value Pr(>F)
## year        6 4.451e+11  7.418e+10   341.8 <2e-16 ***
## county       4 3.186e+12  7.966e+11  3670.2 <2e-16 ***
## Residuals   357 7.749e+10  2.170e+08
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Step 4 & 5: Interpret probabilities and results.

# Our F-Value for year is 341.8 with a corresponding p-value < .05.
```

```
# Our p-value is less than our alpha; therefore, we reject the null
# (H0) and support the alternative (HA). There is a significant
# difference in Average Listing Price across years.
```

```
# Our F-Value for county is 3670.2 with a corresponding p-value <
# .05. Our p-value is less than our alpha; therefore, we reject the
# null (H0) and support the alternative (HA). There is a significant
# difference in Average Listing Price across counties.
```

```
# Step 6: Conduct any Post Hoc Tests if needed.
```

```
TukeyHSD(aov.alp.year.county)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = average_listing_price ~ year + county, data = va.cleaned)
##
## $year
```

	diff	lwr	upr	p adj
2017-2016	8834.117	-934.22656	18602.46	0.1059300
2018-2016	16888.633	7120.29010	26656.98	0.0000100
2019-2016	30664.717	20896.37344	40433.06	0.0000000
2020-2016	55384.247	45588.34834	65180.15	0.0000000
2021-2016	76675.450	66879.55173	86471.35	0.0000000
2022-2016	112579.408	102028.39174	123130.42	0.0000000
2018-2017	8054.517	78.69782	16030.34	0.0460144
2019-2017	21830.600	13854.78115	29806.42	0.0000000
2020-2017	46550.130	38540.58683	54559.67	0.0000000
2021-2017	67841.334	59831.79022	75850.88	0.0000000
2022-2017	103745.292	94828.05511	112662.53	0.0000000
2019-2018	13776.083	5800.26448	21751.90	0.0000103
2020-2018	38495.614	30486.07017	46505.16	0.0000000
2021-2018	59786.817	51777.27356	67796.36	0.0000000
2022-2018	95690.775	86773.53844	104608.01	0.0000000
2020-2019	24719.530	16709.98683	32729.07	0.0000000
2021-2019	46010.734	38001.19022	54020.28	0.0000000
2022-2019	81914.692	72997.45511	90831.93	0.0000000
2021-2020	21291.203	13248.07686	29334.33	0.0000000
2022-2020	57195.161	48247.74802	66142.57	0.0000000
2022-2021	35903.958	26956.54463	44851.37	0.0000000

```
##
## $county
```

	diff	lwr	upr
norfolk city-chesapeake city	-55068.014	-61708.598	-48427.429
portsmouth city-chesapeake city	-158406.338	-165046.922	-151765.753
suffolk city-chesapeake city	-3095.662	-9736.247	3544.922
virginia beach city-chesapeake city	130370.570	123684.030	137057.111
portsmouth city-norfolk city	-103338.324	-109978.909	-96697.740
suffolk city-norfolk city	51972.351	45331.767	58612.936
virginia beach city-norfolk city	185438.584	178752.043	192125.124
suffolk city-portsmouth city	155310.676	148670.091	161951.260
virginia beach city-portsmouth city	288776.908	282090.368	295463.449
virginia beach city-suffolk city	133466.233	126779.692	140152.773

```
##                                p adj
## norfolk city-chesapeake city  0.0000000
## portsmouth city-chesapeake city 0.0000000
## suffolk city-chesapeake city   0.7048552
## virginia beach city-chesapeake city 0.0000000
## portsmouth city-norfolk city   0.0000000
## suffolk city-norfolk city      0.0000000
## virginia beach city-norfolk city 0.0000000
## suffolk city-portsmouth city   0.0000000
## virginia beach city-portsmouth city 0.0000000
## virginia beach city-suffolk city 0.0000000
```

```
# All years are significantly different from each other except:
# 2017-2016. This aligns with the earlier prediction, except for
# 2017-2016.
```

```
# All counties are significantly different from each other except:
# Suffolk and Chesapeake. This aligns with earlier prediction- all
# counties differ significantly, except Suffolk and Chesapeake, which
# are similarly priced.
```

```
#####
```

```
# Test 3 & 4: Differences in Active Listing Count across Years and
# Counties. Step 1:
```

```
# Hypothesis 1:
```

```
# H0 - The Active Listing Count does not differ across years
# (2016-2022).
```

```
# HA - The Active Listing Count does differ across years (2016-2022).
```

```
# Hypothesis 2:
```

```
# H0 - The Active Listing Count does not differ across counties.
```

```
# HA - The Active Listing Count does differ across counties.
```

```
# Step 2 & 3: Compute test-statistic.
```

```
aov.alc.year.county <- aov(active_listing_count ~ year + county, data = va.cleaned)
summary(aov.alc.year.county)
```

```
##           Df    Sum Sq  Mean Sq F value Pr(>F)
## year       6 51564851  8594142   177.7 <2e-16 ***
## county     4 59748027 14937007   308.8 <2e-16 ***
## Residuals 357 17269633    48374
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Step 4 & 5: Interpret probabilities and results.

*# Our F-Value for year is 177.7 with a corresponding p-value < .05.
Our p-value is less than our alpha; therefore, we reject the null
(H0) and support the alternative (HA). There is a significant
difference in Active Listing Count across years.*

*# Our F-Value for county is 308.8 with a corresponding p-value < .05.
Our p-value is less than our alpha; therefore, we reject the null
(H0) and support the alternative (HA). There is a significant
difference in Average Listing Price across counties.*

Step 6: Conduct any Post Hoc Tests if needed.

`TukeyHSD(aov.alc.year.county)`

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = active_listing_count ~ year + county, data = va.cleaned)
##
## $year
##          diff          lwr          upr      p adj
## 2017-2016  -64.0000 -209.8319   81.831861 0.8511086
## 2018-2016 -148.6500 -294.4819  -2.818139 0.0424521
## 2019-2016 -322.7833 -468.6152 -176.951473 0.0000000
## 2020-2016 -722.1627 -868.4059 -575.919477 0.0000000
## 2021-2016 -917.6881 -1063.9314 -771.444901 0.0000000
## 2022-2016 -1017.8500 -1175.3664 -860.333588 0.0000000
## 2018-2017  -84.6500 -203.7212   34.421216 0.3501174
## 2019-2017 -258.7833 -377.8545 -139.712118 0.0000000
## 2020-2017 -658.1627 -777.7374 -538.588022 0.0000000
## 2021-2017 -853.6881 -973.2628 -734.113445 0.0000000
## 2022-2017 -953.8500 -1086.9757 -820.724334 0.0000000
## 2019-2018 -174.1333 -293.2045  -55.062118 0.0003758
## 2020-2018 -573.5127 -693.0874 -453.938022 0.0000000
## 2021-2018 -769.0381 -888.6128 -649.463445 0.0000000
## 2022-2018 -869.2000 -1002.3257 -736.074334 0.0000000
## 2020-2019 -399.3794 -518.9541 -279.804688 0.0000000
## 2021-2019 -594.9048 -714.4795 -475.330112 0.0000000
## 2022-2019 -695.0667 -828.1923 -561.941001 0.0000000
## 2021-2020 -195.5254 -315.6015  -75.449370 0.0000418
## 2022-2020 -295.6873 -429.2635 -162.111111 0.0000000
## 2022-2021 -100.1619 -233.7380   33.414313 0.2855214
##
## $county
##          diff          lwr          upr      p adj
## norfolk city-chesapeake city -136.10811 -235.24557  -36.97064 0.0018174
## portsmouth city-chesapeake city -498.36486 -597.50233 -399.22740 0.0000000
## suffolk city-chesapeake city -461.01351 -560.15098 -361.87605 0.0000000
## virginia beach city-chesapeake city 621.80846 521.98491 721.63200 0.0000000
## portsmouth city-norfolk city -362.25676 -461.39422 -263.11929 0.0000000
## suffolk city-norfolk city -324.90541 -424.04287 -225.76794 0.0000000
## virginia beach city-norfolk city 757.91656 658.09302 857.74011 0.0000000
```



```
## suffolk city-portsmouth city      37.35135  -61.78611  136.48882 0.8399426
## virginia beach city-portsmouth city 1120.17332 1020.34978 1219.99687 0.0000000
## virginia beach city-suffolk city   1082.82197  982.99843 1182.64552 0.0000000
```

```
# All years are significantly different from each other except:
# 2017-2016, 2018-2017, and 2022-2021. This somewhat aligns with the
# prior prediction; however, it was not anticipated that the
# difference between 2017-2016, 2018-2017, and 2022-2021 would be
# insignificant.
```

```
# All counties to be significantly different from each other except:
# Suffolk and Portsmouth. This somewhat aligns with the prior
# prediction. Suffolk and Portsmouth are similar, as predicted.
# Norfolk and Chesapeake are significantly different, in contrast, to
# prior prediction.
```

```
#####
```

```
# Test 5 & 6: Active Listing Count and County Impact on Average
# Listing Price. Step 1:
```

```
# Hypothesis 1: Active Listing Count affects Average Listing Price.
```

```
# H0: The slope of the line in regards to Active Listing Count is
# equal to zero.
```

```
# HA: The slope of the line in regards to Active Listing Count is not
# equal to zero.
```

```
# Hypothesis 2: County affects Average Listing Price.
```

```
# H0: The slope of the line in regards to County is equal to zero.
```

```
# HA: The slope of the line in regards to County is not equal to
# zero.
```

```
# Step 2 & 3: Compute test-statistic.
```

```
options(scipen = 999)
regAll <- lm(average_listing_price ~ active_listing_count + county, data = va.cleaned)
summary(regAll)
```

```
##
## Call:
## lm(formula = average_listing_price ~ active_listing_count + county,
##     data = va.cleaned)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -47157 -13501  -4059   11359   69145
##
```

```
## Coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## (Intercept)    426282.155    3459.216   123.23 <0.0000000000000002
```

```
## active_listing_count      -74.948      2.521  -29.73 <0.0000000000000002
## countynorfolk city      -65268.990    3429.818  -19.03 <0.0000000000000002
## countyportsmouth city   -195757.589    3636.504  -53.83 <0.0000000000000002
## countysuffolk city      -37647.519    3605.060  -10.44 <0.0000000000000002
## countyvirginia beach city 177069.917    3787.295   46.75 <0.0000000000000002
##
## (Intercept)             ***
## active_listing_count     ***
## countynorfolk city       ***
## countyportsmouth city    ***
## countysuffolk city       ***
## countyvirginia beach city ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 20760 on 362 degrees of freedom
## Multiple R-squared:  0.9579, Adjusted R-squared:  0.9574
## F-statistic: 1649 on 5 and 362 DF,  p-value: < 0.0000000000000002

# Check for multicollinearity. No VIF score is above 10.
vif(regAll)
```

```
##                               GVIF Df GVIF^(1/(2*Df))
## active_listing_count 1.896178  1      1.377018
## county               1.896178  4      1.083265
```

Step 4 & 5: Interpret probabilities and results.

*# All variables are significant and none are insignificant.
F-statistic is high with a low p-value ($p < .05$). A one active
listing increase is associated with a \$74.95 decrease in Average
Listing Price. Average Listing Price varied depending on county.
Average Listing Price from highest to lowest is as follows:
Virginia Beach, Chesapeake, Suffolk, Norfolk, and Portsmouth.
Adjusted R-squared is .9574, explaining about 96% of variance,
indicating that this is a strong model.*

Task 7A (10 points)

- Describe the variables you found to be the strongest predictors of price.

*# Of the variables analyzed, county/location was the largest
predictor of price, followed by active listing count.*

*# Using Chesapeake as the referent category: Norfolk was associated
with a \$65,268.99 decrease in price. Portsmouth was associated
with a \$195,757.59 decrease in price. Suffolk was associated with a
\$37,647.52 decrease in price. Virginia Beach was associated with a
\$177,069.92 increase in price.*

A one unit increase in Active Listings was associated with a \$74.95

```
# decrease in price. As active listings increase, average price
# listing decreases.
```

Task 7B (10 points)

- What limitations, biases, or confounding variables could affect the results?

```
# Assumptions: Both ANOVA and linear regression rely on specific
# assumptions, such as normality of residuals and homoscedasticity.
```

```
# Skew and Outliers: Active Listing Count was skewed right to a
# problematic degree. Average Listing Price was also skewed right,
# though not to a problematic degree. Several high and low outliers
# visible in 'Average Listing Price by Year' boxplot. Several high
# outliers present in 'Active Listing Count by Year' boxplot.
```

```
# Confounding variables: Though this analysis identified a
# significant relationship between location and average price
# listing, it did not provide insight into why some locations may be
# evaluated higher than others. This analysis did not include
# socioeconomic factors, such as income, accessibility, crime levels,
# or other amenities that may impact property value. Additionally, it
# did not consider property specific variables like: size, layout,
# age/condition, or property type.
```

```
# Bias: Analysis was very specific to the Hampton Roads area and
# likely would not be helpful in broader/national Real Estate
# analysis.
```